

Nexucon Government Agency Dashboard

Implementation Review Report

Date: August 11, 2026

Scope: Frontend Architecture Mockups & Backend PRD Alignment

Executive Summary:

This report details the implementation processes and architectural updates executed to align the Nexucon Government Dashboard with the comprehensive system reviews. The alignment was conducted across four major review phases, resulting in the creation of 11 new UI dashboards, updates to the global navigation structure, and the complete restructuring of the Backend Product Requirements Document (PRD) to adopt a Microservices architecture.

Phase 1: Digital Eye & BIM Module Mapping

The initial review mapped physical sensor capabilities and 3D modeling features to the dashboard.

Process:

- Analyzed the existing dashboard structure to identify missing components for point cloud processing and anomaly detection.
- Scaffolded new UI routes in Next.js.
- Updated the backend PRD to explicitly include geospatial (PostGIS) requirements.

Pages Implemented:

1. Guided Scan Planning Wizard (digital-eye/scan-planning)
2. Edge Processing & Cloud Pipelines (digital-eye/processing-pipeline)
3. Centralized Issue Management (issues)
4. 4D BIM Progress Validation (bim/progress-validation)

Phase 2: Schedule Compatibility Analysis

This review highlighted critical missing compliance workflows and required government integrations.

Process:

- Created dedicated compliance and enforcement UIs.
- Integrated Nigerian government API connections into the PRD and UI.

Pages Implemented:

5. Stop-Work Order Workflow (inspections/stop-work)

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6. Regulatory Escalation Matrix (regulatory/escalation)
7. Government System Integrations (integrations/government)

Phase 3: Critical Gaps Resolution

The critical gaps review identified 14 distinct missing elements spanning evidentiary compliance, approval workflows, and risk mitigation.

Process:

- Systematically prioritized "Critical" and "High" severity items.
- Scaffolded comprehensive dashboards for auditing, risk assessment, and document validation.

Pages Implemented:

8. Permit Decisions & Delegation of Authority (approvals/decisions)
9. Tamper-Proof Audit Trail (audit/records)
10. Structural Risk Index (analytics/risk)
11. Recurring Offenders & Document Expiry (stakeholders/blacklist & applications/expired)
12. Digitally Stamped BIM Models (bim/approved)
13. FMW Integration Expansion (Federal Ministry of Works added to integrations portal)

Phase 4: Microservices Architecture Fit

The final review demanded a formal mapping of the monolithic applications into highly cohesive microservices.

Process:

- Rewrote docs/GOVERNMENT_BACKEND_PRD.md to classify Django apps as independent services.
- Established specific domains: Digital Eye Services, BIM Services, and Government-Specific Services.

Artifact Implemented:

- Backend PRD Updated: docs/GOVERNMENT_BACKEND_PRD.md

Status: All review objectives successfully accomplished across the UI Mockups and Backend Documentation.